Rainbow domination numbers of hypercubes from covering-code tables

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Abstract

The k-rainbow domination number $\gamma_{rk}(G)$ has been determined for many families of graphs, such as grids, products of cycles and generalized Petersen graphs; we are not aware of values for hypercubes. We point out that, by a theorem of Brešar, Henning and Rall, $\gamma_{r3}(Q_n)$ is the size of an optimal mixed covering code with one ternary and n binary coordinates and radius 1, and, as an immediate observation, $\gamma_{r2}(Q_n)$ is the size of an optimal binary covering code of length $n+1$ and radius 1. Both are tabulated in coding theory. This gives the exact values $\gamma_{r2}(Q_n)$ for $n \le 8$ and $\gamma_{r3}(Q_n)$ for $n \le 6$, and bounds beyond, which we confirm independently for small n by solving the rainbow domination problem directly. For the first open case, $42 \le \gamma_{r3}(Q_7) \le 48$, we show by computer that every 3-rainbow dominating function of Q_7 with weight at most 47 has no non-trivial symmetry.

1 Introduction

Let G be a graph and $k \ge 1$. A *k-rainbow dominating function* is a map $f\colon V(G) \to 2^{\{1,\dots,k\}}$ such that every vertex v with $f(v) = \emptyset$ satisfies $\bigcup_{u \in N(v)} f(u) = \{1,\dots,k\}$. Its weight is $w(f) = \sum_v |f(v)|$, and the *k-rainbow domination number* $\gamma_{rk}(G)$ is the minimum weight. The notion was introduced by Brešar, Henning and Rall [3]; for $k = 1$ it is ordinary domination. Exact values are known for several families, for example $P_3 \,\square\, P_n$ [13], $C_n \,\square\, C_m$ for small n [14, 15], and generalized Petersen graphs [16]. We are not aware of values for hypercubes.

The hypercube Q_n has vertex set $\mathbb{F}_2^n$, with two vertices adjacent when they differ in one coordinate. Its ordinary domination number $\gamma(Q_n)$ is the minimum size $K(n,1)$ of a binary code of length n and covering radius 1, a classical problem in coding theory [4]; $\gamma(Q_9) = 62$ was settled by Östergård and Blass [9]. Domination in hypercubes is surveyed in [6]. Closest to the present note, Azarija, Henning and Klavžar [1] use the fact that Q_{n+1} is the prism $Q_n \,\square\, K_2$ to prove $\gamma_t(Q_{n+1}) = 2\gamma(Q_n)$ and several further domination-type identities for hypercubes. The point of this note is that the rainbow domination numbers of hypercubes are also already in the coding tables.

2 Rainbow domination of Q_n as covering codes

Brešar, Henning and Rall proved that $\gamma_{rk}(G) = \gamma(G \,\square\, K_k)$ for every graph G [3]. For hypercubes we give the correspondence explicitly, since it is what the symmetry result of Section 4 uses.

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Write $K(t, b, 1)$ for the minimum size of a code in $\mathbb{F}_3^t \times \mathbb{F}_2^b$ in which every word is within Hamming distance 1 of a codeword [2, 10]. For $k = 3$, identify a code $C \subseteq \mathbb{F}_3 \times \mathbb{F}_2^n$ with the map

$$f_C(v) = \{a \in \mathbb{F}_3 : (a, v) \in C\}, \qquad v \in \mathbb{F}_2^n,$$

so that $|C| = w(f_C)$, and every map $f\colon \mathbb{F}_2^n \to 2^{\mathbb{F}_3}$ arises from exactly one code.

Proposition 1. *C has covering radius 1 if and only if f_C is a 3-rainbow dominating function of Q_n. Hence $\gamma_{r3}(Q_n) = K(1, n, 1)$.*

Proof. Call $\{(0, v), (1, v), (2, v)\}$ the *column* of v. Changing the ternary coordinate is one step, so a codeword in a column covers all three words of that column. Thus if $f_C(v) \neq \emptyset$, the column of v is covered. If $f_C(v) = \emptyset$, the word (a, v) can only be covered by a codeword (a, u) with u adjacent to v, that is, by some $u \in N(v)$ with $a \in f_C(u)$. So all three words of the column are covered exactly when $\bigcup_{u \in N(v)} f_C(u) = \mathbb{F}_3$, which is the rainbow condition at v. □

For $k = 2$ there is nothing to prove beyond the known identity.

Remark 2 (Observation). By the case $k = 2$ of [3], $\gamma_{r2}(Q_n) = \gamma(Q_n \,\square\, K_2)$, and $Q_n \,\square\, K_2 = Q_{n+1}$. Hence $\gamma_{r2}(Q_n) = \gamma(Q_{n+1}) = K(n + 1, 1)$. This follows at once from [3] and the prism structure used in [1]; we record it only because the resulting values do not seem to have been listed.

3 Values

Table 1 lists the values that follow from the tables of Kéri [7], which collect results from [2, 10, 9, 8] and many other sources. Entries for larger n are bounds as recorded there; later work on lower bounds for $\gamma(Q_n)$, such as [12, 5], may improve some of them.

n	1	2	3	4	5	6	7	8	9	10	11	12
$\gamma_{r2}(Q_n)$	2	2	4	7	12	16	32	62	107–120	180–192	342–380	
$\gamma_{r3}(Q_n)$	2	3	6	8	16	24	42–48	76–84	134–160	253–284	448–548	869–992

Table 1: Rainbow domination numbers of hypercubes, read from the covering-code tables ($\gamma_{r2}(Q_n) = K(n + 1, 1)$, $\gamma_{r3}(Q_n) = K(1, n, 1)$).

Independent check. To guard against errors in the translation, we computed the rainbow domination numbers directly from the graph-theoretic definition, as integer programs solved to optimality with SCIP 10.0 (`src/rainbow.py`). The results, $\gamma_{r2}(Q_n) = 2, 2, 4, 7, 12, 16$ for $n = 1, \ldots, 6$ and $\gamma_{r3}(Q_n) = 2, 3, 6, 8, 16$ for $n = 1, \ldots, 5$, agree with Table 1.

4 The first open case: symmetry

The smallest open value is $42 \leq \gamma_{r3}(Q_7) \leq 48$. The upper bound is attained by giving every vertex of the binary Hamming code of length 7 all three colours: $16 \cdot 3 = 48$. This function is highly symmetric.

The symmetries of the problem are the maps $f \mapsto f^g$ with $f^g(v) = \pi\big(f(\sigma^{-1}v)\big)$, where $\sigma \in \mathrm{Aut}(Q_7)$ and π is a permutation of the three colours. They form the group $\mathrm{Aut}(Q_7) \times S_3$ of order $2^7 \cdot 7! \cdot 6 = 3\,870\,720$. Under the correspondence $C \leftrightarrow f_C$, this is exactly the isometry group of $\mathbb{F}_3 \times \mathbb{F}_2^7$, so symmetries of codes and of rainbow functions match.

Theorem 3. *Every* 3*-rainbow dominating function of* Q_7 *with weight at most* 47 *has trivial stabiliser in* $\mathrm{Aut}(Q_7) \times S_3$.

Method. If the stabiliser of f is non-trivial, it contains an element g of prime order: take any $g_0 \neq 1$ in it, of order m, and a prime $p \mid m$, and set $g = g_0^{m/p}$. Replacing f by an image f^h, which has the same weight, we may assume g is one fixed representative of its conjugacy class. The elements of prime order fall into 46 conjugacy classes (orders 2, 3, 5, 7 with 39, 5, 1, 1 classes). For each representative g, a g-invariant function is constant on the orbits of $\langle g \rangle$, so the least weight of a g-invariant rainbow dominating function is the optimum of an integer program with one variable per orbit of $\langle g \rangle$ on $\mathbb{F}_3 \times \mathbb{F}_2^7$. These 46 programs, with 60 to 288 variables, were solved twice by separately written code: first with HiGHS, and again with SCIP 10.0 on a random conjugate of each representative. In every case the optimum is 48, 54 or 56, never below 48. The full computation, with a per-class table, is in the companion note [11]. □

Both solvers use floating-point arithmetic, so Theorem 3 rests on two independent computations that agree, not on an exact certificate. Its meaning is that any function of weight 47, if one exists, is completely asymmetric; in particular, it cannot be found by searching among symmetric functions, the usual first approach.

5 Remarks

1. The same argument shows that $\gamma_{rk}(Q_n)$ equals the minimum size of a covering code of radius 1 in $A \times \mathbb{F}_2^n$, where A is an alphabet of size k. Tables for some such mixed alphabets also exist [7].

2. Determining $\gamma_{r3}(Q_7)$ is equivalent to determining $K(1, 7, 1)$. A natural approach is the codeword-distribution method used for $\gamma(Q_9) = 62$ [9].

Code and data. `https://github.com/Noah-Parsons/covering-codes`.

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